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Information systems for supply chain management: a systematic literature analysis

Mohammad Daneshvar Kakhki^{a*} and Vidyaranya B. Gargeya^b

^aDepartment of Business Information Systems, Western Michigan University, Kalamazoo, USA ^bDepartment of Information Systems and Supply Chain Management, The University of North Carolina at Greensboro, Greensboro, USA

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Information systems (IS) impact supply chain management (SCM) on processes such as planning, sourcing, and delivering, and at levels ranging from tactical operations to organisational strategy. The vast scope of IS and SCM relationships have resulted in diverse and disintegrated research on the topic. This paper offers a systematic literature analysis at the intersection of supply chain and information systems (SCIS), aiming to provide a classification for existing areas of research. This research is based on an analysis of more than 1500 articles published in peer-reviewed journals over the past four decades to classify topics and methods and to identify major trends and distinguish important research themes. The classification of the literature has identified major clusters of research in SCIS, and suggestions are made for future research in each of the identified clusters. In general, the findings point out that there is a dearth of research on topics such as ‘impact of IT on vertical disintegration of supply chains,’ ‘implications of new technologies for supply chains,’ and ‘concerns related to trust, governance, ownership, privacy, and security of data in supply chains.’ This work provides both researchers and practitioners with an insightful description of the current state of research in SCIS and related future trends in research and practice.

Keywords: Supply chain management; information systems; literature analysis; journal articles; topic clustering

1. Introduction

Organisations in practice, for decades, have been integrating information and communication technologies (ICT) into their production/operations/supply functions for enhancing performance, in terms of reducing costs, improving customer service, and increasing the speed of delivery and reliability. Academic research has also followed suit in studying the inter-connectedness between ICT and production/operations/supply chain management. This has resulted in hundreds of articles being written on the linkage between information systems (IS) and supply chain management. The time is ripe for a systematic analysis in creating a comprehensive framework for studying the two topic and key inter-disciplinary areas.

Supply chain management (SCM) relies heavily on information and communication technologies (ICT) for handling transactions, performing communications, developing management insight, and exchanging information (Gunasekaran and Ngai 2004; Büyüközkan and Göçer 2018). Therefore, the concept of SCM gained attention and was highlighted by researchers only after the introduction of ICT tools (Alfalla-Luque and Medina-Lopez 2009). Use of ICT tools in SCM started in the 1960s with the introduction of electronic data interchange (EDI) systems, followed by material requirements planning (MRP) solutions in the 1970s, manufacturing resource planning (MRP II) in the 1980s, and enterprise resource planning (ERP) in the 1990s (Lavassani, Movahedi, and Kumar 2008; Alfalla-Luque and Medina-Lopez 2009). Numerous ICT tools were developed and adopted throughout the past decades to support the increasingly complex supply chains (Jacobs 2007; Lavassani, Movahedi, and Kumar 2008). Researchers perceived the importance of the use of ICT in SCM and addressed it by developing a growing number of scholarly published papers in scientific journals of various disciplines. The diverse body of this research entails a harmonised taxonomy and an aggregated view of the current state of research to enhance the communication of researchers in different disciplines and predict potential future trends of the research.

The emergence of ICT and how they impact organisations are subject of research in information systems (IS). IS research, not only investigates technological aspects of ICT, but also studies behaviours of individuals and organisations in dealing with evolution, implementation, and adoption of ICT. Consequently, the authors focus on classification of existing literature on supply chain information systems (SCIS), an essential part of SCM (McLaren, Head, and Yuan 2004), to develop an integrative understanding of the current state of SCIS and discuss potential paths in the future of SCIS.

*Corresponding author. Email: mohammad.daneshvarkakhki@wmich.edu

SCIS is 'designed to provide information and information processing capability to support the strategy, operations, management analysis, and decision-making functions' in the supply chain (Tarokh and Soroor 2006). SCIS is a multi-disciplinary topic across several fields, including operations management (OM), IS, computer science, management, and marketing. This diversity has resulted over the years, in discrete research efforts in each discipline. The diverse and fragmented nature of the SCIS research escalates due to the increased pace of research and knowledge production, further disconnecting divergent domains of the literature (Cassell, Denyer, and Tranfield 2006). Therefore, a major concern for SCIS research is the 'difficulty of integration of sub-disciplines' (Tranfield and Starkey 1998, 341). This disconnection leads to the development of literature in SCIS with varying degrees of maturity across various topics in different disciplines. The suggested approach for resolving the concern is literature classification and analysis (Salipante, Notz, and Bigelow 1982). Classification of published papers improves the connection between different domains of research and empowers scholars to position their contribution to the field of research (Salipante, Notz, and Bigelow 1982; Cassell, Denyer, and Tranfield 2006). The integrative role of literature analysis is central in research, and Cassell, Denyer, and Tranfield (2006) articulate that 'the need for a new study is not as great as the need for the assimilation of already existing studies' (216).

The importance of literature analysis is established in both IS and SCM. The SCM knowledge is fragmented across different domains and a systematic review of literature is needed to reveal the gaps in SCM literature (Burgess, Singh, and Koroglu 2006). Likewise, review articles are important for supporting IS and creating the foundation for advancing knowledge (Webster and Watson 2002). Due to that importance, there are scholarly works on classification and analysis of SCM literature (Croom, Romano, and Giannakis 2000; Gunasekaran and Ngai 2004; Burgess, Singh, and Koroglu 2006; Seuring and Gold 2012) and IS literature (Alavi and Carlson 1992; Claver, González, and Llopis 2000; Berthon et al. 2002; Mingers 2003; Chen and Hirschheim 2004; Palvia et al. 2004; Gonzalez, Gasco, and Llopis 2006; Palvia et al. 2015; Paré et al. 2015). Moreover, there are well received published studies at the intersections of the IS and SCM. For instance, the impact of internet of things (IoT) on SCM (Ben-Daya, Hassini, and Bahroun 2018), impact of IT and data analytics on strategic performance of supply chains (Wang et al. 2016; Gunasekaran, Subramanian, and Papadopoulos 2017), role of IT in lean manufacturing (Pinho and Mendes 2017), and latest trends in Industry 4.0 (Liao et al. 2017) are among the very well-received papers that discuss subsets of SCIS. Such highly cited scholarly works show the need for an integrative literature analysis, which supports the development of future research based on the synopsis of the literature (Seuring and Gold 2012). However, and despite this importance, there are limited publications with a holistic view of SCIS that discuss the existing status and trends of the literature. The analysis of SCIS literature is important in the sense that it facilitates future research on SCIS. Also, the high inter-dependence of SCM and IS makes it imperative to know the direction of research and potential emerging topics in the SCIS.

Considering the discussed background about the existing literature, the concern of this research is not only advancing a framework but also providing a taxonomy that is useful for mapping and evaluating the SCIS research. This research aims to reveal patterns, identify gaps, and provide guidelines for the future of research in SCIS. To reach this goal, the authors classify and analyse 1502 papers published in peer-reviewed journals.

Classification and analysis of the SCIS literature contribute to research and has implications for practice. The outcomes of the current research support scholars in both IS and SCM fields and familiarises them with gaps and major trends at the intersection of these two fields. This familiarity improves the relevance of publications through a focus on important and impactful issues. Furthermore, the analysis of the literature and development of a classification framework can provide researchers with a standard taxonomy that enhances the communication between different disciplines. The outcomes are also insightful for practitioners by revealing the potential directions of SCIS development in the future.

The rest of the paper is organised as follows. In the next section, research methodology is discussed, and a classification framework based on the currently published taxonomies in IS, SCM, and subsets of SCIS is developed. Next, results of the literature analysis are presented and discussed. Then, a cluster analysis approach is used to further understand the interaction between various aspects of IS and SCM. Finally, the paper is concluded by discussing results and their implications, deliberating on limitations, and proposing directions for future research.

2. Methodology

This research follows the three-phases approach methodology for literature analysis (Cumbie et al. 2005). In the first phase, a representative pool of papers is collected. Then, the papers are classified based on an appropriate framework in the second phase. Finally, the classified research is reviewed and synthesised in the third phase. The third phase is composed of two steps of general analysis and detailed discussion (Tranfield, Denyer, and Smart 2003). The research method is outlined in Figure 1. The outcomes of these three phases provide a descriptive analysis of the status of research in the SCIS (Tranfield, Denyer, and Smart 2003).

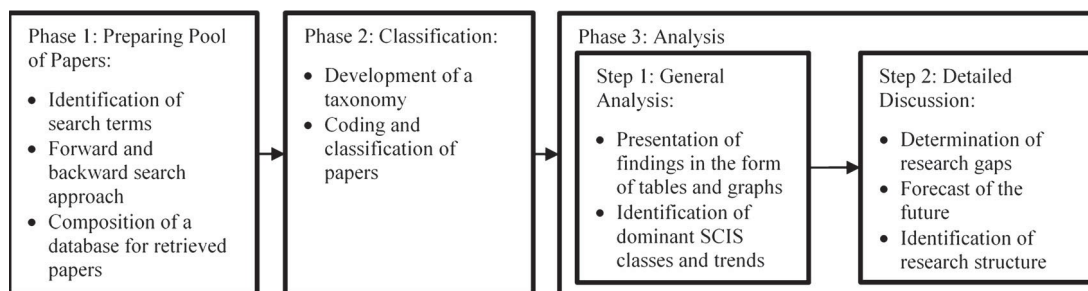


Figure 1. Research method (Cumbie et al. 2005; Adopted from: Tranfield, Denyer, and Smart 2003).

2.1. Phase 1: preparing pool of papers

The current research aims to create a high-quality analysis that is complete and focused on concepts; therefore, the analysis is based on a pool of papers that covers all the relevant literature and is not restricted to specific journals or methodologies (Webster and Watson 2002). To reach this goal, important search terms within SCM and IS (SCM, IS, e-supply chain, etc.) were identified. These search terms were used in combination and separately for retrieval of papers from online scientific databases. The initial search identified several leading and highly cited papers at the intersection of IS and SCM. The inclusion criteria were the use of both IS and SCM search terms or common SCIS search terms in the title or keywords of a paper. Authors enriched their list of search terms going back and forth between identified SCIS papers and extraction of new search terms from the identified SCIS papers for a further search for other SCIS papers. After an initial collection of 100 SCIS papers, authors used an online text mining tool (<http://textalyser.net>) and analysed the titles and abstracts of retrieved papers. This analysis identified one-word, two-words, and three-words phrases occurring most frequently that could be used as search terms. This text analysis was repeated after collection of 500 papers to further enrich the pool of search terms. An abridged list of search terms, assembled for the search process, is presented in Appendix 1. A more comprehensive list is available upon request.

The Webster and Watson (2002) approach for the preparation of a pool of papers was used. The authors conducted an extensive search based on identified search terms in important electronic scientific databases, including Science Direct, Emerald Insight, Web of Science, AIS Electronic Library, Academic Search Premier, Business Source Premier, Taylor & Francis Online, and ACM Digital Library. These scientific databases have been used by other researchers (Gerow et al. 2014). In addition to searching based on search terms, forward and backward search approach have been used based on the highly cited papers to find more relevant papers in SCIS. A manual search in leading IS and OM/SCM journals was conducted to ensure that important articles are not missed from the search results. The metadata of retrieved papers was stored in a database. Considering that publications of a journal may appear in several scientific databases, various queries were run, and manual checks were performed to ensure that each recorded paper is unique in the database.

Overall, 1502 unique papers were retrieved from various databases. The search process and retrieval of papers ended in the first week of September of 2018; therefore, the list of published papers in 2018 is incomplete. Thus, although all the 102 papers published in 2018 are included in analyses and discussions; however, they are excluded from the statistics and charts that are related to the comparison of annual trends. This exclusion is for preventing from comparison biases. Table 1 shows the list of journals that publish SCIS research and number of extracted papers from each. Figure 2 shows the number of SCIS papers published in each year.

2.2. Phase 2: classification

2.2.1. Development of taxonomy

An extensive search in scientific databases was performed and the authors did not find any comprehensive literature classification for SCIS. Therefore, the authors reviewed published classifications in SCM, IS, and subsets of SCIS, including ERP, customer relationship management (CRM), and e-commerce, to develop a suitable classification taxonomy for SCIS. The SCM literature is classified based on content, level of analysis, and methodology (Yusuf, Gunasekaran, and Abthorpe 2004; Burgess, Singh, and Koroglu 2006). IS researchers, compared to a narrower classification in SCM research, employ additional dimensions for classification of IS research. These dimensions are content (topic), method, research model, and research approach (Claver, González, and Llopis 2000; Mingers 2003; Palvia et al. 2004; Gonzalez, Gasco, and Llopis 2006; Palvia et al. 2015).

Table 1. List of journals in a sample of 1502 papers related to SCM and IS.

Journal	Number of papers
International Journal of Production Economics	111
European Journal of Operational Research	61
International Journal of Physical Distribution & Logistics Management	59
Journal of Enterprise Information Management ^a	55
International Journal of Production Research	42
Industrial Management & Data Systems	42
Information & Management	34
Computers & Industrial Engineering	33
Journal of Operations Management	33
Expert Systems with Applications	33
Decision Support Systems	30
Computers in Industry	29
Supply Chain Management: An International Journal	29
The International Journal of Logistics Management	28
Industrial Marketing Management	27
International Journal of Information Management	26
Management Science	20
Production and Operations Management	19
Transportation Research Part E: Logistics and Transportation Review	19
Business Process Management Journal	17
International Journal of Operations & Production Management	17
Journal of Manufacturing Technology Management ^b	16
Journal of Cleaner Production	16
Production Planning & Control	14
Journal of Business & Industrial Marketing	12
Benchmarking: An International Journal	12
MIS Quarterly	12
The Journal of Strategic Information Systems	12
Omega	11
International Journal of Information Systems and Supply Chain Management	11
Other Journals (294 journals)	622

^aThe article count for the 'Journal of Enterprise Information Management' includes published papers in 'Logistics Information Management' and 'Logistics World,' which are two earlier names of the journal.

^bThe article count for the 'Journal of Manufacturing Technology Management' includes published papers in 'Integrated Manufacturing Systems' journal, which is the earlier name of the journal.

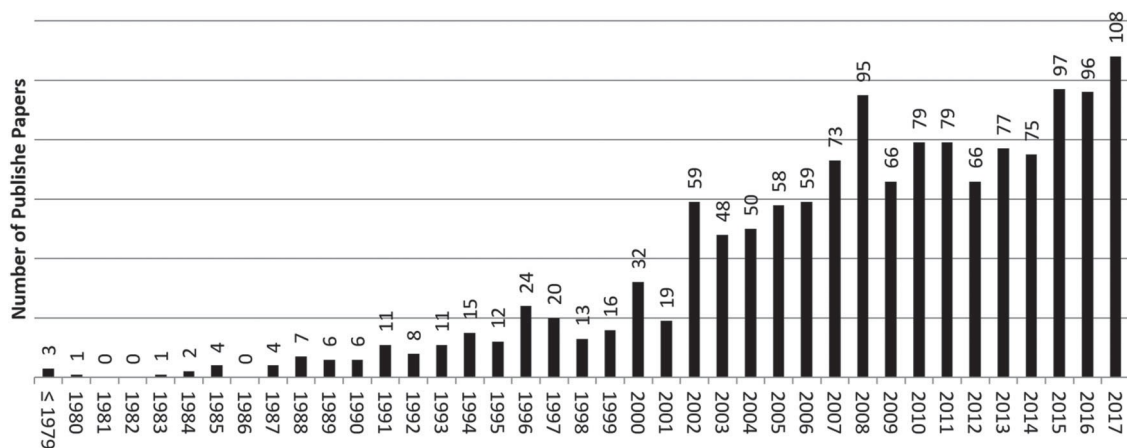


Figure 2. Number of published SCIS papers in each year.

There are published studies that classify subsets of SCIS, including ERP, e-commerce, and CRM. The employed taxonomies in these studies borrow their classification dimensions from SCM or IS literature. The frameworks for ERP classification suggest classes based on research topic (Aloini, Dulmin, and Mininno 2007; Moon 2007; Rerup Schlichter

and Kraemmergaard 2010), methodology of research (Aloini, Dulmin, and Mininno 2007; Rerup Schlichter and Kraemmergaard 2010), and the research discipline that the paper is from (Rerup Schlichter and Kraemmergaard 2010). Furthermore, the literature on e-commerce is classified based on the research topic (Ngai and Wat 2002). Finally, frameworks for CRM classify the literature based on topic (Ngai 2005; Wahlberg, Strandberg, and Sandberg 2009).

Each of the classification frameworks classified literature in SCM, IS, or SCIS from a specific angle. Since the current research aims to provide an integrated view of these different research areas under the comprehensive title of SCIS. The

Table 2. Dimension of classification of supply chain and IS.

Domain	Classification criteria	References
Research Method	Design science, field research, field/laboratory experiment, frameworks and conceptual models, literature review/analysis, mathematical modelling, qualitative research, secondary data, speculation/commentary, survey.	(Mingers 2003; Palvia et al. 2004)
Topics (SCM)	- Strategic management: strategic networks, control in the supply chain, time-based strategy, strategic sourcing, vertical disintegration, make or buy decisions, core competencies focus, supply network design, strategic alliances, strategic supplier segmentation, world class manufacturing, strategic supplier selection, global strategy, capability development, strategic purchasing. - Relationships/partnerships: relationships development, supplier development, strategic supplier selection, partnership sourcing, supplier involvement, supply/distribution base integration, supplier assessment (ISO), guest engineering concept, design for manufacturing, mergers acquisitions, joint ventures, strategic alliances, contract view, trust, commitment, partnership performances, relationship marketing. - Logistics: integration of materials and information flows, MRP, waste removal, vendor managed inventory (VMI), physical distribution, cross docking, logistics postponement, capacity planning, forecast information management, distribution channel management, planning and control of materials flow. - Best practices: just in time (JIT), MRP, MRP II, continuous improvement, tiered supplier partnerships, supplier associations (Kyoryoku Kai), leverage learning network, quick response, time compression, process mapping, waste removal, physically efficient vs. market-oriented supply chains. - Marketing: relationship marketing, internet supply chains, customer service management, efficient consumer response, efficient replenishment, after sales service. - Organisational behaviour: communication, human resources management, employees' relationships, organisational structure, power in relationships, organisational culture, organisational learning, technology transfer, knowledge transfer.	(Croom, Romano, and Giannakis 2000) To make the list more representative of the current research, few new topics were added to the original list
Topics (IS)	- Decision support: Big data, business intelligence/data analytics/expert system, decision support system & executive IS, group support systems. - IT technologies: blockchain, cloud computing, cryptocurrency, databases, end user computing, hardware, internet, internet of things (IoT), media and communications, mobile computing, social media and social computing, social networks, software and programming languages, telecommunications and networking. - IS-enabled business: business process, e-government, electronic commerce /business, health information technology, knowledge management, organisational design, SCM, virtual teams. - Enterprise systems: customer relationship management (CRM), enterprise resource planning (ERP), interorganisational systems (IOS). - Environment of IT: internal or external environment of IT, global information technology (GIT), innovation, IS education, IS research, IS staffing, IT and culture, outsourcing and offshoring, project management, societal issues, sustainability. - IS development and application: IS design and development, IS evaluation, IS functional applications, IS implementation, IS management and planning, IS usage/adoption, IT value, security and privacy.	(Palvia et al. 2015; Palvia et al. 2017) To make the list more representative of the current research, few new topics were added to the original list

authors decided to incorporate three classification dimensions, namely: SCM topic, IS topic, and methodology. The incorporation of these three dimensions yields enough detail that is expected for classifications and prevents from high-level clustering of SCIS in the classification (Hassini, McLaren, and Vuong 2008). The three dimensions of the framework that is developed for classification are presented in Table 2. These dimensions are used for coding classification of retrieved SCIS papers.

2.2.2. Coding

All the 1502 retrieved papers have been reviewed and coded using the proposed taxonomy. Accordingly, all papers were coded considering the research method, IS topic, and supply chain topic. The authors coded each paper based on its title, keywords, abstract, and for most cases, the main body of the paper. This procedure has been used in similar scholarly works (Philip and Pedersen 1997; Lu, Huang, and Heng 2006; Zhu and Augenbroe 2006).

The coding process followed the method introduced by Palvia et al. (2015). Coders discussed and went through the definitions of the classification dimensions in several meetings prior to starting the coding. The goal of the meetings was to create a uniform understanding of the classification and purpose of the research and resolve ambiguities. For the first round of coding, 50 papers were assigned to two coders: one with a Ph.D. in IS and is a co-author of the paper, and one with a Ph.D. in operations management and a post-doctoral fellow researcher in SCM. In this process, the inter-coder reliability was calculated to be 68% (Weber 1990). Since the inter-coder reliability was under 0.80, the coders had another set of meetings to discuss differences in their coding and resolve disagreements. In the second round of coding, another 50 papers were assigned to coders and the inter-coder reliability was calculated to be 83% which is an acceptable level of agreement between coders based on the literature. The rest of the papers were coded by one researcher, the co-author of the paper. To ensure that the reliability is at an appropriate level, authors followed the insider-outsider method used by Gerow et al. (2014). In this method, 30 papers were randomly selected and assigned to the second researcher for coding. The new inter-coder reliability was 89%. The inconsistencies were resolved, and the coding was updated to eliminate the determined inconsistencies.

3. Results

The results of the analysis of each dimension are presented in this section. Furthermore, the most important trends and issues are discussed. These results are presented in form of tables and graphs, making them easier to follow.

3.1. Methodology trends

Table 3 presents the trend in research methodology in SCIS. A considerable share of SCIS papers in IS and OM journals employ mathematical modelling and survey. Qualitative research, design science, and frameworks and conceptual models are other important methods with high rank in SCIS research. The diversity of research methods in SCIS shows the healthy characteristics of the SCIS research (Flint et al. 2012). The trends in Table 3 show that the majority of SCIS research is based on explanatory methods such as mathematical modelling, design science, frameworks, and conceptual models. In recent years, the field witnessed more published confirmatory research and the relative share of speculator/ commentary and qualitative research is decreasing. Beside confirmatory research, the number of published literature analyses increased in recent years. These results show that the earlier dominant inductive research approach is gradually changing and there are more publications with a deductive approach. The subsequent research uses more systematic methods of data collection and analysis. These trends are signs of a shift in the SCIS research paradigm and show that researchers are trying to give a distinctive identity to SCIS by synthesising existing research and developing related theories.

3.2. Topics, usage and trends

Major topics and number of their appearance in retrieved papers are presented in Table 4 (Some papers have more than one topic in either IS or SCM). Among the supply chain topics, 'supply/distribution base integration' ranks first and appears in 20% of papers. The second most important topic in the supply chain is 'planning and control of material flows,' which accounts for 11% of the papers. This topic discusses inventory replenishment, bullwhip effect, forecasting, tracking, etc. The topic of 'supply chain performance' ranks third with 9% of the papers. This topic, which is highly related to the business value of IT literature, contains empirical studies that measure the impact of different IS related constructs on various supply chain performance aspects (Trkman et al. 2010; Oliveira, McCormack, and Trkman 2012; Daneshvar Kakhki and Palvia 2016). Supply chain performance is usually used as a dependent variable jointly with different IS related constructs such as

Table 3. Trend of research methodology in SCIS.

	≤ 1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	Total	% Total
Mathematical Modelling	13	1	4	3	1	5	1	17	5	8	8	16	17	17	18	23	18	21	19	20	24	22	34	315	22%
Survey	8	2	4	1	1	2	1	8	6	14	12	13	15	18	14	20	22	18	19	23	25	27	28	301	21%
Qualitative Research	18	6	2	2	6	9	4	11	14	8	12	12	18	25	9	9	11	8	7	12	11	14	13	241	17%
Design Science	10	7	3	1	3	10	3	9	12	11	3	7	6	13	7	8	9	10	10	6	15	11	15	189	13%
Frameworks and Conceptual Model	16	4	1	1	5	3	4	9	8	5	7	7	4	5	11	4	6	4	14	7	13	15	9	162	11%
Speculation/commentary	22	1	2	3	0	2	5	2	4	2	3	1	1	8	1	2	5	0	2	1	4	1	2	74	5%
Literature	0	1	2	0	0	1	1	2	0	0	7	0	7	5	3	8	0	4	1	4	5	4	6	61	4%
Review/Analysis	4	2	0	1	0	0	0	0	2	2	5	1	3	5	6	3	2	4	2	2	2	2	3	51	4%
Field Research	0	0	2	0	0	1	0	1	1	0	3	0	4	1	0	2	5	2	3	2	1	2	3	33	2%
Secondary Data	0	0	0	0	0	0	0	1	0	2	1	2	0	0	0	0	2	0	0	0	0	0	0	8	1%
Field/Laboratory Experiment	91	24	20	12	16	33	19	60	52	52	61	59	75	97	69	79	80	71	77	77	100	98	113	1435	
Total																									

^aThe number of 1435 (total identified research methodologies) does not include papers published in 2018, which are 112 different methodologies in 102 published studies.

ISTopics																					
SCM Topics	Business Intelligence and Analytics					IS Topics					Other Topics										
	Information Sharing	Business Intelligence and Analytics	IS Usage/Adoption	IS Design and Development	Electronic Commerce/Business	IS Functional Applications	Internet of Things	Knowledge Management	IT Value	IS Management and Planning	IS Implementation	IS Evaluation	Electronic Data Interchange	Security and Privacy	Collaboration IS	Business Process Management	Customer Relationship Management	Social Networks	Other Topics	Total	
Supply/Distribution Base Integration	75	7	81	19	28	17	10	4	13	11	9	9	3	13	3	8	5	7	2	338	
Planning and Control of Materials Flow	34	33	4	10	13	6	19	12	5	13	6	7	7	6					2	181	
Supply Chain Performance	26	13	9	28	5	13	5	8	5	10	5	5	6	1	1	1	5		2	154	
Integrated Physical and Information Flows	10	2	15	9	30	17	10	18	2		12	2	3	1	1	2	1			148	
Distribution Channel Management	4	39	1	8	9	18	19	6	1	3	2	3	1	1			1			110	
Physical Distribution		13	1	2	5	19	9	2	1	1		2	1					2		59	
Supply Network Design	1	24	1		8	5		1	1	2				1	1					48	
Supply Chain Technologies	1			5	6	3	4	18	2			2								47	
Contract View, Trust, Commitment	10	2	5		1	3		2		1	1			3	14	1		1		47	
Green Supply Chain	6	8		9	5	3	1		1	3	1	2								39	
Technology Transfer		1		26	1							1								30	
Relationship Development	3	1	5	3	4	2	2	3	2	3	1							1		30	
Forecast Information Management	5	5	2	5		5			2	3	1			1						29	
JIT, MRP, MRP II	6	4		3	2	1	2	2	1		2		3							26	
Supplier Involvement	1	2	9	3	2	2	1		2		1		2		3					26	
Supply Chain Visibility	1	1			4	1	4	4				3	2		1					22	
Control in the Supply Chain	1	2		3	1	1	2	6						1						21	
Supply Chain Risk		5		3	1	1	1	2	1				1	2				1		18	
Partnership Sourcing	3		1	3	1	7														17	
Relationship Marketing		4	1	1	1	1	1		3								3			16	
Relationships Development	2		3		1	1	1		2		3		2							15	
Other Topics																				233	
Total	200	198	164	156	144	140	102	93	60	60	54	46	35	32	27	17	15	14	13	79	1649

investment in technologies, IS adoption, information sharing, coordination, and integration. Most papers that discuss ‘supply chain performance’ use an empirical method such as survey, experiment, or secondary data as a methodological approach for their research. ‘Integration of information and material flows’ in supply chain holds the fourth rank and appears in 9% of the papers. The ‘distribution channel management’ topic with 7% of papers is the fifth most important topic related to SCM.

The first rank in IS topics with 12% of the total is ‘information sharing.’ The relatively high number of research articles in this IS topic matches the need of supply chain for ‘supply/distributor-based integration.’ Another 12% of the papers discuss ‘business intelligence and analytics (BI&A).’ BI&A includes different pieces of research related to various generations of decision support, business intelligence, and data analytics that are used for extracting information from regular data and big data. BI&A is discussed in different domains, including logistics, supplier selection, marketing decision support, and general decision support systems. In addition, this topic includes different technologies for decision support, including group decision making, simulation, and various data analytics approaches. The topic of ‘interorganisational systems (IOS)’ ranks third in IS and 10% of published papers are focused on this topic. IOS include papers in the field of business to business IS, business to customer IS, purchasing IS, and other collaborative interorganisational IS. Topics of ‘IS Usage and adoption’ and ‘IS design and development’ rank fourth and fifth with nearly nine percent of published papers on each of the two topics.

Table 4 also demonstrates the intersection of IS and SCM topics. Results show that papers related to ‘interorganisational systems’ and ‘supply/distribution base integration’ have the highest co-appearance among all published papers and 81 SCIS papers that account for 5% of all retrieved papers are in this intersection of SCIS topics. The second most discussed topic is related to ‘information sharing’ and ‘supply/distribution base integration’ with 75 papers and nearly 4.5% of all papers. The intersection between ‘business intelligence and analytics’ and ‘distribution channel management,’ which covers the planning and design of distribution channels, ranks third in the table.

Figure 3 shows change and development of various SCIS topics over time. The size of the bubbles in the chart is an indication of the volume of research in that period. The intersection of ‘Supply/distribution integration’ topic and two IS topics of ‘interorganisational systems’ and ‘information sharing’ are the most discussed topics in SCIS. As it is shown in Figure 3, these two SCIS topics have been major trends in the research since the beginning and they are still important in the

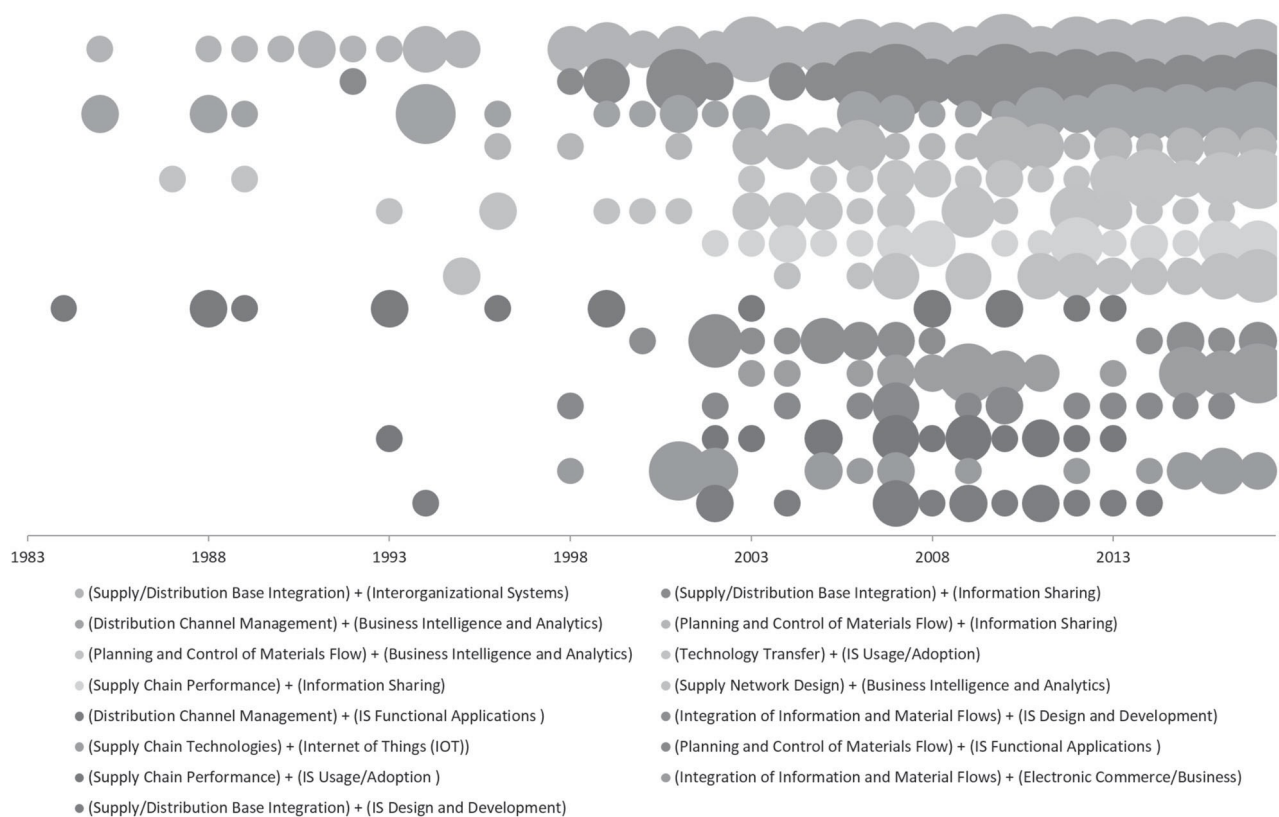


Figure 3. Trend in SCIS research topics.

SCIS. Another important topic is BI&A in distribution channel management. This topic has considerable variance, while in some years many papers were published on the topic, but the topic was neglected in other years. However, in the past few years, researchers focused on the topic and there is a considerable number of published studies discussing the topic.

As presented in Figure 3, some topics are relatively new and still have growth potential. Among them, application of new technologies in SCM and implementation of BI&A in control of material flow are two important topic trends. Some research topics have been discussed in the field from the beginning and researchers are still working on them. 'IS usage and adoption,' and 'IS functional applications' are two examples of this group. There are other topics such as 'IS design and development' for 'integration of information and material flows' that received high attention from researchers in SCIS in earlier years, but the number of papers in this topic is decreasing in recent years.

4. Discussion and analysis of topics

An in-depth analysis of the interrelationships between different topics in the fields of IS and SCM, which is depicted in Table 4, requires sophisticated data analysis techniques. This research employs clustering techniques to create an enhanced understanding of interactions between the two fields.

4.1. Topology based clustering

The intersection of topics from IS and SCM could be viewed in the form of a graph with topics being nodes and relationships (co-appearance of topics in a paper) being edges. Nodes related to two topics are linked together if both topics appear in one paper. For example, the work of Zhao, Xie, and Zhang (2002), titled 'the impact of information sharing and ordering co-ordination on supply chain performance,' is related to the 'supply chain performance' topic in SCM and the 'information sharing' topic in IS. This paper creates two linked nodes (topics) in the graph. Forming this graph is useful and enables researchers to benefit from effective tools for filtering and clustering the data to find key topics and their links and to reveal important patterns in the topic analysis (Shneiderman and Dunne 2012). The high volume of published works in SCIS creates a complex graph. Since the graph consists of many topics (nodes), clustering simplifies it and enables the discovery of important patterns. Using clustering techniques along with visualising data enhances the analysts' abilities to explore the data (Perer and Shneiderman 2009).

There are various techniques for simplification of graphs through determination of clusters. These clustering techniques are being used in literature analysis. For instance, clustering is used for co-citation analysis studies in IS (Culnan 1986; Walstrom and Leonard 2000) and SCM (Pilkington and Liston-Heyes 1999; Charvet, Cooper, and Gardner 2008; Pilkington and Meredith 2009). The use of clustering techniques in such bibliographic research facilitates classification and analysis of large pools of papers from different perspectives (Ding 2011).

Clustering algorithms identify clusters based on the topology of the graphs (topology-based clustering). Clustering in a network of topics use algorithms that divide graphs into different sub-graphs and then measure of the quality of created clusters (Clauset, Newman, and Moore 2004). Each of these algorithms aims to create sub-clusters with a larger internal connection between nodes compared to their external connection (Kumar, Novak, and Tomkins 2010). Use of clustering algorithms for identification of clusters at the intersection of IS and SCM facilitates analysis of SCIS by presenting it in different levels of abstraction.

Ding (2011) discussed that the outcome of the cluster identification of the Clauset–Newman–Moore clustering algorithm is similar to the Girvan–Newman algorithm, which is the most commonly used clustering approach. However, the Clauset–Newman–Moore clustering algorithm is a more stable method that yields consistent and robust results. Therefore, the authors used the Clauset–Newman–Moore clustering algorithm to create clusters of relevant topics. The result of topology-based clustering of SCIS papers is presented in Figure 4. To simplify the graph, the relationships between the topics in the same group are shown through direct lines. The width of each line is representative of the strength of the link between the two topics. The relationship between topics in different clusters is shown in an aggregated format (grey wide lines in the background the connect clusters). The size of each bubble is an indication of the volume of related research.

The clustering algorithm, which is executed using NodeXL software package, identifies six clusters of topics. A name was given to each of these clusters based on their topic contents. Figure 4 shows the identified clusters and their names. The list of topics in each cluster is presented in Appendix 2. As it is shown in Figure 4, the presented clusters, which on an average are composed of 15 topics, are highly inter-related. While some topics have a higher degree of centrality and are highly inter-connected with topics in other clusters, some other topics are isolated in their clusters and do not have a connection with topics in other clusters. Each of these clusters is discussed in the following sections.

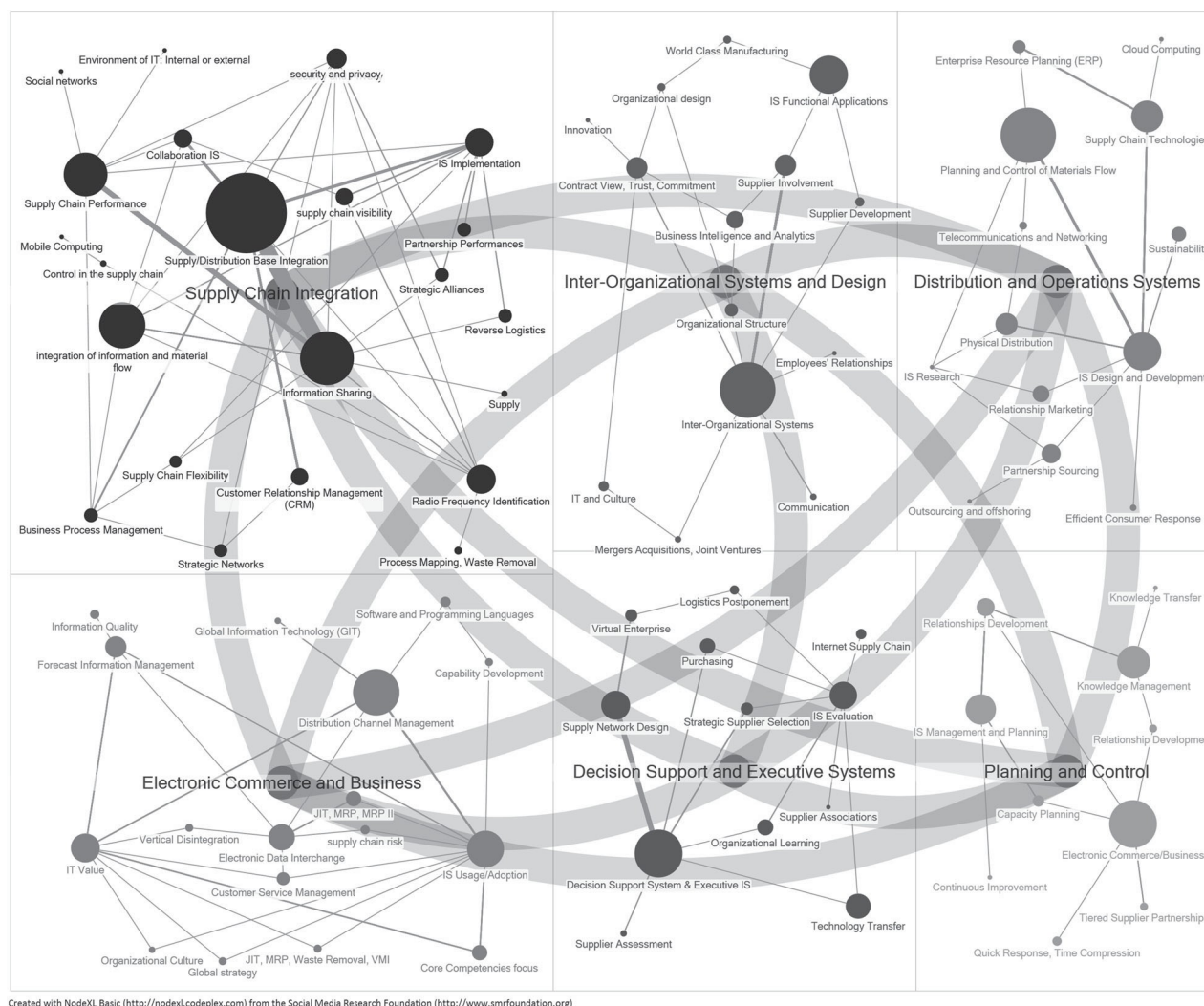


Figure 4. Clustering topics into 6 groups using Clauset-Newman-Moore cluster algorithm (NodeXL is used for clustering and developing the graph).

4.2. Clusters of research in information systems in supply chain management

4.2.1. Supply chain integration

Supply chain integration is discussed in SCIS literature at different levels of strategic alliance and collaboration, operational cooperation and integration, and technical integration. Studies at all these levels identify the important role of integration in performance; however, each level discusses this importance from a specific angle.

At the strategic level, supply chain integration is focused on employment of IS tools for better coordination of supply chain partners for value creation (Daneshvar Kakhki 2018). The traditional view discusses value as a product or service that is created by discrete efforts of suppliers (Bettencourt, Lusch, and Vargo 2014), whereas more recent literature considers value as an entity that is co-created in a partnership between supply chain players and customers (Frohlich and Westbrook 2001; Xie et al. 2016). The co-creation of value is about the employment of the mutual resources and capabilities to serve customers, and recent advances in social media and big data analytics facilitate communication between customers and supply chain partners towards co-creation of value (Erevelles, Fukawa, and Swayne 2016; Maglio and Lim 2016).

IS-enabled integration at the strategic level is an enhancement of financial performance and competitive advantage; however, integration at tactical and operational levels seeks to improve the operational performance. There are published studies that investigate role of integration and communication on various aspects of business, including enhancement of coordination, improvement in visibility, and improvement of supply chain processes (Zhao and Xie 2002; Huo, Zhang, and Zhao 2015; Şahin and Topal 2018).

Beside the positivist studies that analyse and theorise IS-enabling of supply chains and its impact on performance, there is interpretivist research that uses qualitative study approach and design science methods for explaining the nature of integration. The interpretivist research in this cluster reports on successful IS design and development and IS usage and adoption projects cases and proposes design archetypes that are appropriate for integration. For instance, Kulvatunyou and Wysk (2000) proposed an enterprise integration approach to improve the integration of supply chain processes. Rai, Patnayakuni, and Seth (2006) observed that the performance improvement could not be achieved by merely being equipped with ICT tools; rather it is necessary to create an integrated ICT platform and ICT-enabled processes. Creation of such an infrastructure requires standardised architecture of ICT solutions within the firm and rationalised data architecture compatible with global standards that enable the firm to communicate with its environment (Ross 2003). An important and emerging topic in this area is Industry 4.0 (Liao et al. 2017), which through an employment of enhanced communication standards and methods, leads to a higher level of integration in the supply chain.

Overall, the majority of studies suggest that IS-enabled supply chain integration leads to higher levels of performance. The impact of supply chain integration on the performance of firms is reflected in the work of Frohlich and Westbrook (2001) who investigated 322 manufacturers and their integration strategies. They analysed the level of integration with both suppliers and customers through analysis of integrative activities, including shared information and joint data interchange. Based on the analysis, the Frohlich and Westbrook (2001) concluded that manufacturing firms with higher levels of supplier and customer integration will have higher business performance.

Although the positive impact of the integration on the performance of supply chains is established, integration leads to concerns such as the creation of inertia in supply chain partners, which negatively impacts flexibility. The investment associated with IS for supply chain integration adds to the challenge by creating a barrier for deserting an existing relationship or forming new relationships. There are studies on the development of flexible and modular ICT components for facilitation of integration (Daneshvar Kakhki, Nemati, and Hassanzadeh 2018). Also, there is research on standardisation of IS tools and communication methods that facilitate integration. While existing studies address IS-enabled supply chain integration, the literature still needs more clarification on managing the risks and benefits of the topic.

Information sharing and challenges related to privacy and security are other concerns related to supply chain integration. Some of the published studies related to this cluster discussed the role of data sharing through the entire supply chain (Dejonckheere et al. 2004) and some deliberated on the importance of data sharing with supply side, customer side, or across logistical processes (e.g. Gavirneni, Kapuscinski, and Tayur 1999; Kim and Umanath 2005; Sahin and Powell Robinson 2005). This information sharing, despite its potential benefits, is limited by trust issues where companies are concerned about the privacy and security of their information (Wang, Ye, and Tan 2014).

Within the mutual relationship of supply chain partners, information becomes an important asset for the entire supply chain. Appropriate sharing of information and controlled access to it is vital for supply chains to sustain their competitive advantage in the long term. An analysis of different aspects of information security in supply chains shows that all technical, formal and informal security aspects affect the performance in both inter and intra-organisational settings (Sindhuja 2014). Besides the traditional networks and methods of information sharing that are the cause of security and privacy challenges, the emergence of IoT and the addition of such hardware devices to supply networks add to the security challenge. However, and despite the concerns and potential negative impacts of information sharing, there are limited studies that discuss the empirical evidence related to trust, privacy, and security in the governance and information sharing aspects of supply chain integration. Therefore, a detailed examination of information security in supply chains, the effect of leakage, and the mechanisms through which managers can enhance their supply chain security still merits further attention.

4.2.2. *Electronic commerce and business*

E-commerce facilitates business transactions through computer-based networks and is discussed from different perspectives, including communication, business, service, and online perspectives (Kalakota and Whinston 1997). The online and communication perspectives define the e-commerce as an infrastructure that facilitates transactions and enables buying and selling products and information online. From the service perspective, e-commerce is defined as a tool that serves suppliers and customers with lower costs and better quality at a higher speed. With the availability of the online infrastructure and considering the lower costs of transactions, there are opportunities for innovative changes in supply chains. These changes empower the business perspective of e-commerce by reducing transaction costs and enhancing coordination. This cluster contains a good body of research on how different online, communication, service, and business perspectives of e-commerce improve the performance of supply chains by facilitating transactions at lower costs.

There are studies that address how e-commerce impacts capability development of supply chain partners, and lead to new and innovative businesses (e.g. Cassivi, Léger, and Hadaya 2005). Enhanced communication and reduced transaction

cost are drivers of such new and innovative business opportunities. Accordingly, the evolution of businesses as a result of the adoption of e-commerce is shifting the market structure from traditionally vertical form to a more horizontal form (Robey, Im, and Wareham 2008). The horizontal market structure enables direct connection between buyers and suppliers, which results in increasingly complex supply networks. Accordingly, there are studies that address the shift in the market structure by investigating how e-commerce leads to vertical disintegration (e.g. Evangelista and Sweeney 2009).

Currently, there are studies on the transformative role of e-commerce; however, there are limited studies that address the complex nature of e-commerce enabled supply networks. Moreover, the more horizontal forms of market structure entail interactions between companies and customers in complex networks, rather than supply chains. However, most of the reviewed papers in this cluster are focused on dyadic or chain level interactions. Therefore, the study of complex network interactions caused by e-commerce is suggested.

4.2.3. *Interorganisational systems*

IOS are shared IS in dyadic or chain levels. These systems serve the purpose of improved information flow between organisations and facilitate the integration of supply chains. IOS provide supply chain partners with an improved linkage to other upstream and downstream chain members and enhance their responsiveness to varying needs and demands of their market. While the topic of IOS could be studied under the cluster of e-commerce, the clustering algorithm proposed a separate cluster for IOS. This may perhaps happen due to the large number of published papers that are merely focused on this topic. This cluster deals with design and development, implementation, and usage and adoption of IOS for various needs of supply chain and is highly interconnected to both 'supply chain integration' and 'electronic commerce and business' clusters.

IOS was introduced by Barrett and Konsynski (1982). Since then, and for many years, the major focus of IS researchers was EDI. In later years, the research on this topic expanded and encompassed new generations of technologies such as the internet and open standard technologies. The IOS related papers can be categorised in one of the three mainstreams discussed by Robey, Im, and Wareham (2008), namely: adoption of IOIS, the impact of IOS on governance, and organisational consequences of IOS adoption. IOS are discussed in the literature based on their characteristics and their impact on supply chain operations (Saeed, Malhotra, and Grover 2011). The performance implications of IOS are discussed in distribution channel management, innovative supply chain practices such as VMI and JIT, vertical disintegration, and improved supply chain coordination.

The general trend and future of the research in this cluster is affected by two major shifts. First, the EDI and the proprietary technologies that were used in the IOS are being replaced by internet-based and open standard solutions. The proprietary technologies resulted in the high costs and unwanted consequences associated with the formation and termination of a buyer-supplier relationship. The internet-based open standard solutions made the termination and initiation of buyer-supplier relationships easier. The ease in the establishment of new relationships through more standard IOS makes companies increase their external connections. As a result, the second shift in the IOS context is the increasing complexity of supply networks resulted from increasing direct buyer-supplier relationships. Considering these major shifts, the IOS research will face a number of social and economic challenges related to ownership and governance in expanded supply networks (Robey, Im, and Wareham 2008). These challenges demand further attention and investigation by researchers.

4.2.4. *Planning and control*

Planning for and control of supply and demand are the means of controlling physical flow in supply chains (Jacobs 2011). Therefore, planning requires enhanced visibility and traceability of a supply chain. Accordingly, researchers address supply chain visibility through innovative IT-enabled supply chain tracking and control. Moreover, published studies discuss the use of BI&A tools and their role in the analysis of supply chain data, which improve the preciseness of demand forecasting and quality of planning. While the use of new technologies for an enhanced planning and control are addressed in the literature, there are challenges and issues for today's business environment that demand further attention.

First, the business environment is extremely volatile due to intense competition and constant changes in customer needs. Dealing with volatility, beside incorporation of suitable and responsive analytical tools for developing plans and controlling them, requires careful design of organisational structures and flexible governance of interorganisational relationships. In other words, organisational capabilities such as flexibility and agility, alignment, and adaptability are important dependent variables that need to be addressed in this cluster of research.

Second, the new threats that are imposed by environmental changes and social inequality should be addressed in SCM. Accordingly, responsible sourcing, closed loop supply chains, product lifecycle management, and reverse logistics are among the topics that researchers should incorporate in planning and control of supply chains. While there are notable studies on the IT-enablement of the supply chain for addressing the sustainability (Dao, Langella, and Carbo 2011), the topic is still underdeveloped and demands further attention.

The third potential line for enhancement of this cluster of research is the level at which research is done. Based on the discussion on the importance of network view in the previous sections, it is important to discuss planning and control at the network level. Yet, the focus of the majority of papers related to this cluster is on dyadic and chain levels.

4.2.5. *Business intelligence and analytics*

SCM is becoming increasingly complex due to factors such as varying customer demands, rapid technological changes, and increasing competitive pressure. On the other hand, the operation of these complex supply chains creates large volumes of data. Since IT-enabled decision making enhances the managerial abilities and reduces the difficulty of SCM (Sambamurthy, Bharadwaj, and Grover 2003), the available data coupled with new data analytics tools enable supply chain managers to deal with the complexity and to enhance the performance of these supply chains. Therefore, firms are heavily investing in their capabilities for data analysis by incorporating decision support tools, executive systems, and BI&A technologies (e.g. Kappelman et al. 2016).

Papers related to the BI&A discuss topics such as design and development, implementation, usage and adoption, and IT value of BI&A tools in different domains of a supply chain, including network design, supplier relationship, customer relationship, supplier development, planning, and supply chain operations. There is published research based on design science and case study methods that discuss the development and use of decision support tools in supply chains. Also, there are empirical studies that justify the positive impact of investing in decision support tools on the performance of supply chains. Besides the current accumulated body of knowledge, researchers suggest that there are issues for proper employment and use of BI&A tools in supply chains. Here, some of these topics are discussed.

The first challenge is about our understanding of how BI&A impact supply chains. As statements in the two prior paragraphs imply, it is important to study the mechanisms through which BI&A impact SCM. Furthermore, it is insightful to investigate the success factors affecting the implementation of BI&A tools in the interorganisational settings. Also, it is essential to study the performance implications of BI&A in SCM. Previous studies on implementation of IS and IT business value are mainly focused within the boundaries of firms.

The second challenge is related to the modification of the dependent variable in the business value of IT studies. The topic of co-creation of value, as discussed in the 'supply chain integration' cluster, is an emerging topic with high practical importance. The literature suggests that the business value of IT research should be focused on value creation (Kohli and Grover 2008). Accordingly, researchers employ evidence from case studies to show how BI&A tools contribute to value co-creation (Tan and Zhan 2017), and there are a number of published frameworks on the role of BI&A in value co-creation (Maglio and Lim 2016; Lenka, Parida, and Wincent 2017; Sheng, Amankwah-Amoah, and Wang 2017). However, and despite the initial conceptualisation and provided anecdotal evidence, there is a limited empirical research on the topic.

The third challenge is related to employment and focus of BI&A tools. In the age of big data, the firms access to information is beyond its analytical capability. Therefore, even a firm with a high level of analytical capabilities 'needs to target its analytical efforts where they will do the most good' (Davenport, Harris, and Morison 2010, 73). Accordingly, it is important for organisations to use their analytical resources for detecting problems with strategic value. Therefore, proper problem sensing focus is critical for efficient use of BI&A resources (Overby, Bharadwaj, and Sambamurthy 2006; Daneshvar Kakhki 2018). However, there is limited research on the topic.

Finally, the newness of the BI&A topic creates challenges. The topic is relatively new, and the technology infrastructure is increasingly changing. This provides new research opportunities. For instance, there is limited research on the proper investment and selection of appropriate BI&A tools and technologies for dealing with constant changes of the technology. Further, proper employment of the technology for dealing with the complexity and dynamism of business landscape needs attention. Furthermore, the concept of Industry 4.0, which is highly intertwined with BI&A, is getting an increasing attention from researchers and practitioners. But, existing literature is limited on how Industry 4.0 impacts supply chains. The last, but not the least, recommendation for future research in this cluster is the knowledge creation process and its related issues, including privacy and security, ownership and governance.

4.2.6. *Distribution and operations systems*

SCM is dealing with a managerial conundrum of achieving higher levels of customer responsiveness and reducing costs of the supply chain. In response to this challenge, researchers are addressing the role of ICT in cost reduction, in service level enhancement, and in developing innovative approaches to handle the conundrum. Accordingly, published research has a wide range and includes various analytical models, design propositions, qualitative, and empirical studies. Moreover, these studies appear at different levels of analysis, including internal analysis, dyadic level, and chain level. Despite the

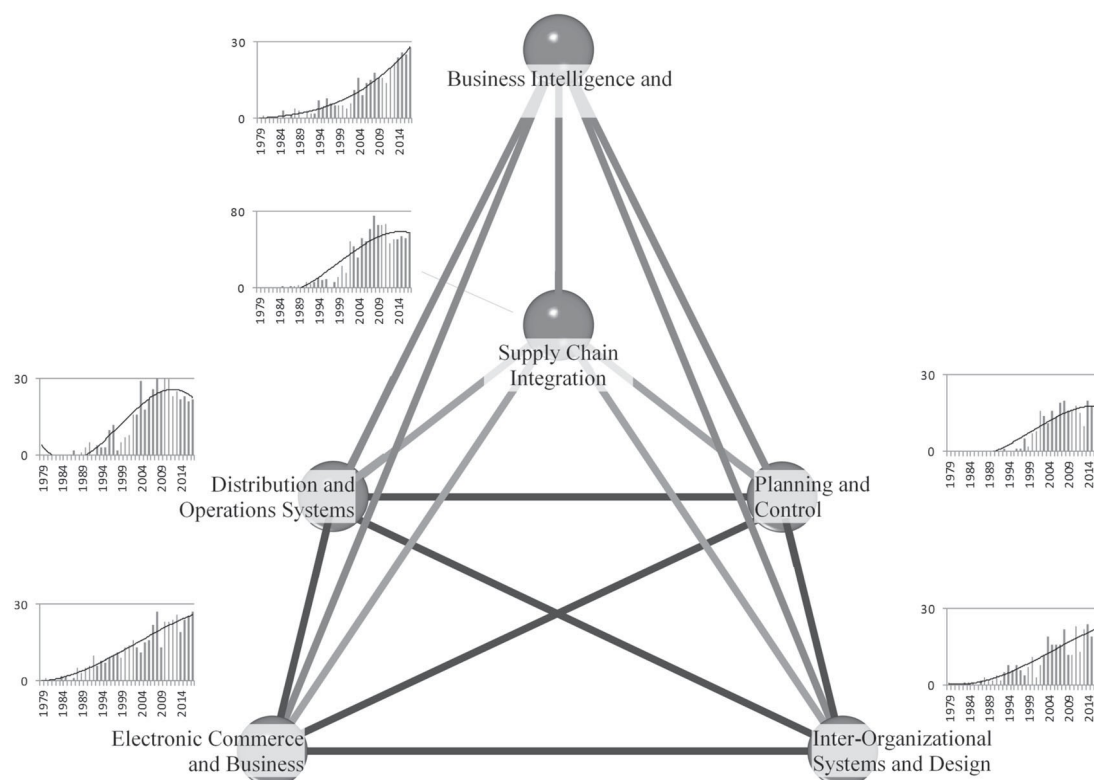


Figure 5. Structure and trend of research in SCIS.

diversity of methods and levels of analysis, and due to the nature of research in this cluster, the majority of studies develop an analytical model for developing a response to achieve local improvements and optimums (Saad and Bahadori 2018).

The bulk of research in this cluster is focused on the internal or dyadic level and very little research is published at the chain and network levels. These papers focus on improved intra-organisational performance by proposing improvements in technology use, enhancement of learning capabilities, and renovation of organisational design. Such studies are focused on problems such as buyer–supplier relationships, supplier assessment, purchasing, and technology transfer. Two shifts seem necessary in this area. First, with the availability of big data, and access to data science tools, it is possible to incorporate additional variables to the existing variables being used in the analyses. An example would be an incorporation of weather data in dynamic planning for logistics operation (Steinker, Hoberg, and Thonemann 2017). Achieving this requires incorporation of a mix of dynamic programming and data science tools for problem modelling purposes. Second, considering the discussed trends in the ‘supply chain integration’ and ‘electronic commerce and business’ clusters, research should address distribution and operations problems at the chain and network levels.

4.3. SCIS research structure

To conclude and summarise the discussion, the authors present the relationship between the six identified clusters in Figure 5. A time dimension is added to each cluster to show how the existing body of knowledge is growing. Figure 5 shows two pyramids in which one of them is inside the other one with the same base. The base consists of ‘planning and control,’ ‘electronic commerce and business,’ ‘interorganisational systems,’ and ‘distribution and operations systems’ in its four corners. The external pyramid has the ‘business intelligence and analytics’ at the apex and the internal pyramid consists of ‘supply chain integration’ at the apex. Each corner is accompanied by a graph showing the research trend in the cluster. The vertical axis of the graph is the number of published topics and the horizontal axis is the year of publication. The trend line is a polynomial with an order of 3, which shows the direction of knowledge accumulation and development in each of the clusters.

As topics trends in Figure 5 show, the slopes of research trend in ‘distribution and operations systems,’ ‘supply chain integration,’ and ‘planning and control’ are becoming flat. The majority of published papers are in supply chain integration, but the rapid expansion of research in the topic seems to slow down. While the research trend for these three clusters is slowing down, the trend for ‘electronic commerce and business’ and ‘interorganisational systems’ is increasing. The slope

of the trend line is positive, and the number of published papers is increasing at a constant rate. The fastest growth belongs to 'business intelligence and analytics,' which also has a positive slope and an increasing number of published papers. In the coming years, research will be more focused on the electronic solutions for commerce and data transfer and application of decision tools in the management of supply chains and supply networks.

4.4. Contributions and implication

The current research provides researchers with numerous opportunities for future research. First, developing a taxonomy for classification of SCIS papers is insightful and could be further used by researchers for similar studies that are interdisciplinary and aim to classify existing research at the intersection of two fields. Second, the classification of publications based on topics and methodologies represent the current state of research in SCIS. Moreover, a graphical representation of results furnishes researchers with an understanding of the existing trends and gaps in SCIS research. Accordingly, the current research grants the researchers a good perception of the combination of topics and methods that are used in the SCIS research. Finally, the third contribution of this research is the incorporation of the clustering technique for classification and discussion of the existing body of knowledge and proposition of potential path to future research. This holds value for both new and established researchers. New researchers can adjust their line of research by incorporating the provided insight gained from this paper. More established researchers, reviewers, and journal editors can use the findings of this research to tutor their graduate students, to inform and strategise direction of their journals and call for papers (e.g. Palvia and Daneshvar Kakhki 2016).

Findings of this research are also insightful for practitioners. Entrepreneurs and business owners can direct their resource investment and capability development activities based on the explained relationship between important topics of SCIS and based on the discussed future of IT-enabled supply chain improvements that are provided in this research. Moreover, active practitioners in the field of SCM can use the provided insight for developing their personal capabilities and preparing themselves for future changes in the market.

4.5. Limitations

Current research classifies available published studies about SCIS in reviewed journals. As a result, published papers in high profile journals are analysed along with published papers in journals with a lower quality. Therefore, there is a wide range of difference between classified papers. Comparing papers from journals with different audiences, research rigour, and approaches might lead to bias in research results and discussions. To prevent the bias in the discussion section, authors try to provide support for their arguments from well-established research works published in the OM/SCM or IS area.

Another limitation of this work is the potential flaws in the search process. Authors spent a considerable amount of time in the search process finding SCIS papers. Despite their effort, and due to the scope of work, some papers might be neglected accidentally or because of the errors in the search process. For instance, in the initial search process, the authors did not search for the 'industry 4.0' search term. As a result, several papers were missed. Later, and in the revision of the paper, authors included some of the missed search terms that were noted by reviewers or identified in complementary text analyses. Since the search process has been designed in accordance with the literature and 'no stone was left unturned' in the search process, the authors of this research believe that such mistakes are minimal and have a negligible impact on the results of this study.

Finally, the third limitation is related to the dilemma about the number of clusters. On the one hand, a higher number of clusters enables a more detailed analysis. On the other hand, the higher number of clusters reduce the parsimony and focus of research. Therefore, authors considered two objectives to make the clustering more meaningful: (1) presentation of enough detail in clusters, and (2) maintaining the parsimony in presented results to enable detection of important research areas. The authors intentionally did not set a limit for a number of clusters to be determined by the clustering algorithm. The six identified clusters may seem insufficient to explain all important clusters of SCIS. However, working with six clusters makes is easier to present, visualise and discuss the results. Moreover, and based on the available data and retrieved papers, focusing on a limited number of clusters enable a clearer distinction of emerging new topics in the field.

5. Conclusion

The current research develops and executes a taxonomy for classification of the literature at the intersection of IS and SCM. Authors conducted a comprehensive search process to retrieve related published papers and coded the papers based on their topics and methodology. The results show trends of knowledge accumulation and methods of developing SCIS knowledge. Moreover, authors employed a clustering technique and classified topics related to SCIS into six clusters of

'supply chain integration,' 'electronic commerce and business,' 'interorganisational systems,' 'distribution and operations systems,' 'business intelligence and analytics,' and 'planning and control.' Following the identification of research trends, methods, and important clusters of research, authors provide a discussion on the prevailing body of knowledge, existing gaps, and potential directions for future research.

There are important considerations and trends in SCIS studies that this paper suggests future research should focus on. First, rather than focusing on the impact of ICT on the performance of supply chains, authors suggest that future research should study how ICT enabled supply chains can co-create value with their customers. Second, the IOS and e-commerce tools are moving towards open standard and internet-based technologies. This trend has had a significant impact on the cost of implementation and adoption process and merits further investigation. Third, the new platforms for communication between supply chain partners may pose security and privacy challenges to customer data that should be addressed in future studies. The fourth consideration should be the need for appropriate usage of huge volumes of data which is becoming more and more accessible for supply chain management. The adoption and application of big data analytics and other business intelligence tools are creating a new opportunity for supply chain managers to leverage the performance. A detailed analysis of the perquisites and consequences of business intelligence tools in SCM needs further thought and investigation. The fifth consideration is the use of IS tools for the preparation of supply chains for dealing with environmental challenges and inequality issues that are important for sustainable development. Finally, the growth of new technologies including IoT, 3D printing, virtual reality, self-driving cars, and drones and advent of Industry 4.0 is rarely discussed in the literature. The future research should investigate different aspects of these technologies and their impact on supply chains.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendix 1. A sample list of used searched terms for retrieving SCIS papers

A more comprehensive list is available upon request.

Domain	Keywords
SCM	Bullwhip effect, distribution, green supply chain management, intelligent transportation system, inter-functional coordination, interorganisational compatibility, interorganisational relationships, inventory, just-in-time, logistics, logistics transportation, multimodal logistics, purchasing, retailing chains, reverse logistics, route selection strategies, SCM, supplier management, supply chain, supply chain agility, supply chain capabilities, supply chain competence, supply chain cooperation, supply chain management, supply chain management practices, supply chain performance, supply chain strategy, supply network, transportation, transportation equipment, transportation planning, vehicle routing problem
IS	Analytics, big data, business intelligence, data models, blockchain, cryptocurrencies, data sharing, decision support systems, distributed systems, e-business technologies, e-commerce service platform, e-commerce technologies, electronic commerce, electronic data interchange (EDI), geographic information systems (GIS), information acquisition, information and communication technologies (ICT), information feedback, information flow, information integration, information privacy, information processing, information networks, information security, information sharing, information systems, information systems strategy, information technology, information technology adoption, information technology alignment, information visibility, internet of things (IoT), IT competence, knowledge creation, knowledge transfer, management information system, public information service, social media, radio frequency identification, system-to-system integration, web, web services
SCIS	B2B, CRM systems, e-commerce, customer integration, customer interface management, customer knowledge, customer-oriented information system, e-Logistics information system, e-supply chain, interorganisational information systems (IOIS), interorganisational systems (IOS), IOIS/IOS integration, logistics information systems, logistics decision support system (DSS), logistics integration, supply chain application integration, supply chain integration, supply chain visibility, supplier integration, supply chain information support, supply chain integration, supply chain simulators, supply chain software, supply chain decision support system (DSS), web-based supply chain applications

Appendix 2. List of topics in each cluster

Cluster	SCM topics	IS topics
Supply Chain Integration	• Control in the supply chain • Integration of information and material flow • Partnership performances • Process mapping, waste removal • Reverse logistics • Contract view, trust, commitment • Strategic alliances • Strategic networks • Supply chain flexibility • Supply chain performance • Supply chain visibility • Supply/distribution base integration	• Security and privacy • IS implementation • Business process management • Customer relationship management (CRM) • Information sharing • Collaboration IS • Mobile computing • Radio frequency identification (RFID) • Environment of IT: internal or external • Social networks
Electronic Commerce and Business	• Capability development • Core competencies focus • Customer service management • Distribution channel management • Forecast information management • Global strategy • JIT, MRP, MRP II, waste Removal, VMI • Organisational culture • Supply chain risk • Vertical disintegration	• Electronic data interchange • Electronic commerce/business • IS usage/adoption • IT value • Global information technology (GIT) • Software and programming languages • Information quality
Interorganisational Systems	• Communication • Contract view, trust, commitment • Employees' relationships • Mergers acquisitions, joint ventures • Organisational structure • Supplier development • Supplier involvement • World class manufacturing	• IS functional applications • Interorganisational systems • Business intelligence and analytics (BI&A) • IT and culture • Organisational design • Innovation
Distribution and Operations Systems	• Efficient consumer response • Partnership sourcing • Physical distribution • Planning and control of materials flow • Relationship marketing • Supply chain technologies • Sustainability	• IS research • IS design and development • Enterprise resource planning (ERP) • Telecommunications and networking • Cloud computing • Outsourcing and offshoring
Business Intelligence and Analytics	• Internet supply chain • Logistics postponement • Organisational learning • Purchasing • Strategic supplier selection • Supplier assessment • Supplier associations • Supply network design • Technology transfer	• Virtual enterprise • Business intelligence • Data analytics • Decision support system & executive IS • Knowledge management
Planning and Control	• Capacity planning • Continuous improvement • Knowledge transfer • Quick response, time compression • Relationship development • Tiered supplier partnerships	• IS management and planning • IS evaluation